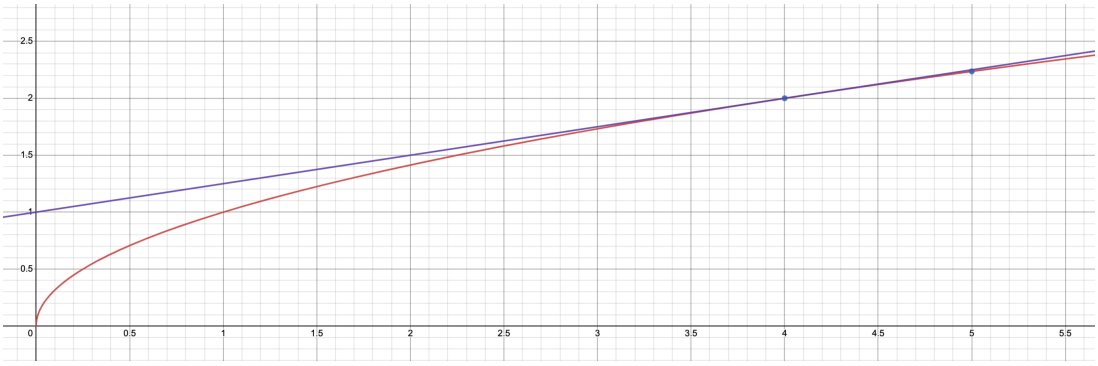
Linear Approximation

Suppose we are asked to approximate $\sqrt{5}$ without a calculator. Note that, near the point $(4,2)$, the graph of $y=\sqrt{x}$ looks like the tangent line to the graph at $(4,2)$.



Since $f'(x)=\frac{1}{2\sqrt{x}}$, the slope of the tangent line at 4 is $f'(4)=\frac{1}{2\sqrt{4}}=\frac{1}{4}$, and the equation of the tangent line is therefore $y=2+\frac{1}{4}(x-4)$. Thus we approximate

$$\sqrt{5}\approx 2+\frac{1}{4}(5-4)=2+\frac{1}{4}=2.25$$

Note that, since the graph is concave down, it lies below the tangent line, so 2.25 is an overestimate. The exact value is $\sqrt{5}=2.236\ldots$, so the error in this approximation is less than 0.014. This appoximation is called the *tangent line approximation* or *linear approximation*.

1.) Use the tangent line approximation to estimate the following:

(a) $\sqrt{9.1}$

(b) $\sin(62^\circ)$

(c) $(1.01)^3$ (*Hint:* Use the linear approximation of $f(x) = (1+x)^n$ at $x = 0$.)

Summary. Suppose we wish to approximate $f(x)$ for some number x. Choose the closest number a to x such that $f(a)$ is known. If f is differentiable at a, then $f(x) \approx L(x) = f(a) + f'(a)(x-a)$. The following proposition shows that this approximation becomes arbitrarily good by taking x sufficiently close to a.

Proposition. If f is differentiable at a, then $\lim_{x\to a}(f(x) - L(x)) = 0$.

2.) (Optional) Prove the proposition above. (It is just a rewriting of the definition of $f'(a)$).

Differentials

We have just seen how to use the linear approximation to estimate function values. It can also be used to estimate the change in the value of the function corresponding to a small change in input. Let f be differentiable at a. If we change a to $x = a + \Delta x$, then the change in the function is

$$\Delta y = f(a + \Delta x) - f(a).$$

3.) To distinguish it from Δy, denote the change in the tangent line approximation by dy. If we also denote Δx by dx, use the tangent line approximation to show that, under a change $x \mapsto x + dx$ in input, we have

$$\boxed{\Delta y \approx dy = f'(x)dx}.$$

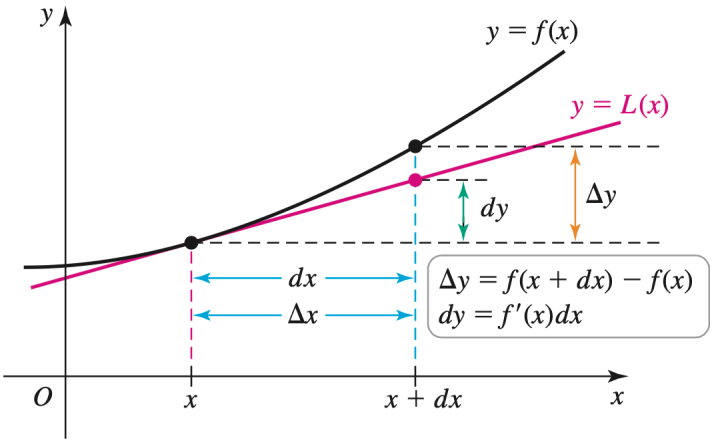


Figure 1: The change in $y = f(x)$ resulting from a change dx in x is approximately $dy = f'(x)dx$.

Terminology. The quantities dy and dx are called *differentials*. One thinks of dx as an independent variable, denoting a "small" change Δx in x, while dy is a dependent variable, approximating the change Δy in the function by $dy = f'(x)dx$. Note also that if $dx \neq 0$, then division by dx gives

$$\frac{dy}{dx} = f'(x),$$

which agrees with our previous notation for the derivative. The concept of differentials therefore gives independent meaning to the quantities dx and dy.

4.) For each of the following functions, find dy and use it to estimate Δy when if we change $x = 3$ to 3.1.

(a) $y = x^2 + 2x$

(b) $y = \cos(x)$

(c) $y = e^{x^2}$